

ACS880 Drive Troubleshooting Guide

ACS880 Drive Troubleshooting Guide (sample excerpt)

This excerpt is for pipeline testing only. It mimics the structure of an industrial drive manual: safety notes, a fault-code table, and a wiring diagram placeholder.

Safety

Disconnect mains and wait five minutes for DC bus capacitors to discharge before measuring terminals. Use a meter rated for the drive voltage class.

Do not defeat the Safe Torque Off circuit while diagnosing a trip.

Symptom: drive trips on overvoltage during deceleration

1. Confirm the fault code on the control panel is 3210 DC overvoltage.
2. Check deceleration time. Increase ramp-down if the load is high inertia.
3. Inspect the brake resistor and chopper if a braking option is installed.
4. Measure DC bus voltage while stopping. Compare to the trip threshold in the hardware manual for this frame size.
5. If the resistor is open, replace it and retest with a reduced speed reference before returning the machine to production.

Fault codes

Fault code reference

Code	Name	Typical cause	First check
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3210	DC overvoltage	Decel too fast / resistor open	Ramp, brake resistor
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3220	DC undervoltage	Supply dip or loose input	Mains, fuses, wiring
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2310	Overcurrent	Short, mechanical jam	Motor cables, load
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4210	IGBT overtemperature	Blocked airflow or high ambient	Fan, heatsink, cabinet
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5090	STO active	STO inputs open	STO wiring, safety relay
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7510	Communication timeout	Fieldbus cable or terminator	Bus, node ID, baud
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Fault 3210 procedure

Detailed procedure for fault 3210 DC overvoltage

Context

The DC bus rises when the motor regenerates into the drive faster than the bus can absorb. High-inertia loads and short deceleration times are the usual combination. A failed brake resistor looks the same on the keypad.

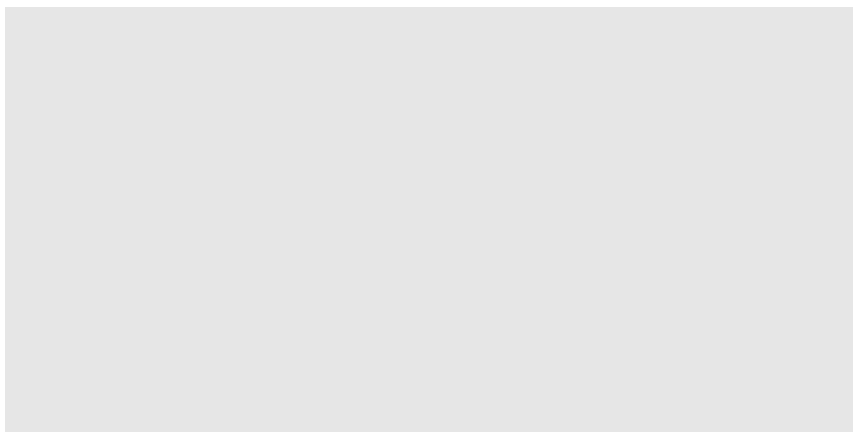
Checks in order

Measure input voltage on L1 L2 L3 under load. If the supply is already high, fix that first. Then lengthen parameter 22.02 deceleration time by 50 percent and repeat the stop. If the trip remains, inspect the brake chopper and resistor with power isolated. An open resistor reads infinite ohms at the chopper terminals.

If the trip only happens on emergency stop, the ramp is being bypassed. Confirm that emergency-stop handling is coast-to-stop or uses a dedicated braking path, not a faster ramp into an already high bus.

Figure 4-2 Brake chopper connections

Typical R+ / R- brake resistor terminals on the power unit. Keep leads short and twisted. Do not share this cable with STO wiring.



R+ resistor R- (frame R4 shown)

FIELD SERVICE NOTE (scanned)

Brake resistor measured 12.4 ohm. Replace if greater than 15 ohm.

STO jumpers intact. Fan intake blocked with dust & cleaned.